

Test Plan for Conditional Add Logic (add_decision)

DUT is purely combinational. Inputs: A[8:0], M[7:0], Q_lsb Outputs: sum = A + {1'b0, M}, do_add = Q_lsb

DUT functionality to be tested	Type of stimulus	Expected behav for DUT
Sum computation	A=0,M=0 ; A=0,M=nonzero ; A=nonzero,M=0 ; A=nonzero,M=nonzero	sum = A + M
do_add follows Q_lsb	Q_lsb=0 then Q_lsb=1 with same A,M	do_add = Q_lsb, sum unchanged
Carry / 9-bit width	A=0FF M=FF ; A=1FF M=01	full 9-bit sum, wraps if >9 bits
Independence of sum from Q_lsb	hold A,M, toggle Q_lsb	sum constant, only do_add changes
Corner cases	A,M = {min,max} combinations	sum = A+M, wrap if >9 bits

Note: max sum = 511+255 = 766, exceeds 9 bits and wraps to 0xFE.